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ELECTRODE FOR SECONDARY BATTERY, SECONDARY BATTERY PROVIDED WITH SAME, AND METHOD FOR MANUFACTURING ELECTRODE FOR SECONDARY BATTERY

Abstract

The present disclosure provides an electrode for a secondary battery capable of measuring the width of a first mixture layer and the width of a second mixture layer when the first mixture layer and the second mixture layer are stacked and formed on the surface of a metal foil. A positive electrode body (electrode for a secondary battery) **301** includes a strip-like positive electrode foil **301a** and a positive electrode mixture layer **301b** provided on the positive electrode foil **301a**. The positive electrode mixture layer **301b** includes a first positive electrode mixture layer **311** provided on the positive electrode foil **301a** and a second positive electrode mixture layer **312** provided on the first positive electrode mixture layer **311**. The first positive electrode mixture layer **311** has a width greater than a width of the second positive electrode mixture layer **312**.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/766,938, filed on Apr. 6, 2025, which is a Bypass Continuation Application of PCT International Application No. PCT/JP2020/043635, filed on Nov. 24, 2020, in the WIPO, which claims priority to Japanese Application No. 2019-213861, filed on Nov. 27, 2019, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to an electrode for a secondary battery having a mixture layer provided on the surface of a metal foil, a secondary battery provided with the same, and a method for manufacturing an electrode for a secondary battery.

BACKGROUND ART

[0003] Conventionally, automobile manufactures have developed electric vehicles (EV) and hybrid electric vehicles (HEV) that use an electric motor to assist part of the drive, and as a power source for those vehicles, there is a demand for a high-capacity, high-output secondary battery. As such a secondary battery, there is known a secondary battery including a positive electrode having a positive electrode mixture layer provided on the surface of a positive electrode metal foil and a negative electrode having a negative electrode mixture layer provided on the surface of a negative electrode metal foil. There is also known an electrode for a secondary battery in which a positive electrode mixture layer and a negative electrode mixture layer are each formed by stacking two different types of mixture layers in order to improve the characteristics of the secondary battery (see, for example, Patent Literature 1).

[0004] Patent Literature 1 discloses a negative electrode for a lithium ion secondary battery including a current collector foil and a negative electrode mixture layer, in which the negative electrode mixture layer includes a first negative electrode mixture layer and a second negative electrode mixture layer, the first negative electrode mixture layer is disposed on the current collector foil, and the second negative electrode mixture layer is disposed on the first negative electrode mixture layer.

CITATION LIST

Patent Literature

[0005] Patent Literature 1: JP 2018-181539 A.

SUMMARY OF INVENTION

Technical Problem

[0006] By the way, when two mixture layers are stacked and formed on the surface of the metal foil as disclosed in the aforementioned Patent Literature 1, the widths of the two mixture layers may affect the battery characteristics, and thus it is required to entirely inspect the widths of the two mixture layers.

[0007] However, when the two mixture layers are stacked and formed on the surface of the metal foil, mixture slurries are usually applied by using coating dies that have the same shape and size. Consequently, a lower (metal-foil side) mixture slurry layer is covered by an upper mixture slurry layer, and after a drying step, a lower (metal-foil side) mixture layer is covered by an upper mixture layer. This makes it difficult to measure the width of the lower mixture layer, and thus it is required to first apply and dry the lower mixture slurry and measure the width of the lower mixture layer, and then apply and dry the upper mixture slurry and measure the width of the upper mixture layer. That is, the width of the lower mixture layer and the width of the upper mixture layer cannot be measured simultaneously. Unfortunately, this makes the manufacturing process more complicated.

[0008] The present invention has been made in view of the foregoing, and provides an electrode for a secondary battery, a secondary battery provided with the same, and a method for manufacturing an electrode for a secondary battery capable of measuring the width of a first mixture layer and the width of a second mixture layer simultaneously when the first mixture layer and the second mixture layer are stacked and formed on the surface of a metal foil.

Solution to Problem

[0009] In view of the foregoing, the electrode for a secondary battery according to the present invention is an electrode for a secondary battery including a strip-like metal foil and a mixture layer provided on the metal foil, in which the mixture layer includes a first mixture layer provided on the metal foil and a second mixture layer provided on the first mixture layer, and the first mixture layer has a width greater than a width of the second mixture layer.

Advantageous Effects of Invention

[0010] According to the present invention, it is possible to provide an electrode for a secondary battery, a secondary battery provided with the same, and a method for manufacturing an electrode for a secondary battery capable of measuring the width of a first mixture layer and the width of a second mixture layer simultaneously when the first mixture layer and the second mixture layer are stacked and formed on the surface of a metal foil.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is an external perspective view of a rectangular secondary battery including a positive electrode body (electrode for a secondary battery) according to one embodiment of the present invention.

[0012] FIG. 2 is an exploded perspective view of a rectangular secondary battery including a positive electrode body according to one embodiment of the present invention.

[0013] FIG. 3 is a perspective view of the electrode group shown in FIG. 2, which is partially expanded.

[0014] FIG. 4 is a cross-sectional view taken along line 100-100 of FIG. 3.

[0015] FIG. 5 is a plan view of a positive electrode body according to one embodiment of the present invention.

[0016] FIG. 6 is a flow chart showing one example of the step of preparing a positive electrode body according to one embodiment of the present invention.

[0017] FIG. 7 is a schematic configuration view of a coating/drying apparatus that is used in a coating step S3, a drying step S4, and an inspecting step S5

[0018] FIG. 8 shows a structure around a die head of the coating/drying apparatus.

[0019] FIG. 9 shows a structure of a die head of the coating/drying apparatus.

[0020] FIG. 10 is a cross-sectional view schematically showing the state where a first positive electrode mixture slurry layer and a second positive electrode mixture slurry layer are formed on a positive electrode foil.

[0021] FIG. 11 is a schematic view around an inspecting device.

[0022] FIG. 12 is a schematic view around an inspecting device according to a modification of the present invention.

DESCRIPTION OF EMBODIMENTS

[0023] Hereinafter, a secondary battery including an electrode for a secondary battery according to an embodiment of the present invention will be described.

[0024] A rectangular secondary battery 20 including a positive electrode body (electrode for a secondary battery) 301 according to one embodiment of the present invention will be described with reference to FIG. 1 to FIG. 3. Herein, an example of applying the rectangular secondary battery 20 to a lithium ion secondary battery will be described.

[0025] FIG. 1 is an external perspective view of the rectangular secondary battery 20 including the positive electrode body 301 according to one embodiment of the present invention. FIG. 2 is an exploded perspective view of the rectangular secondary battery 20 including the positive electrode body 301 according to one embodiment of the present invention. FIG. 3 is a perspective view of an electrode group 3 shown in FIG. 2, which is partially expanded. As shown in FIG. 1 and FIG. 2, the rectangular secondary battery 20 includes a battery case 1 and a lid 6. The battery case 1 houses the electrode group 3 serving as a power generator, and the upper opening of the battery case 1 is sealed with the lid 6. The lid 6 is welded to the battery case 1 by laser welding. The battery case 1 and the lid 6 form a battery container.

[0026] The lid 6 is provided with a positive electrode external terminal 8A and a negative electrode external terminal 8B. Via the positive electrode external terminal 8A and the negative electrode external terminal 8B, the electrode group 3 is charged, and electric power is supplied to an external load. The lid 6 is provided with a gas discharge valve 10. The gas discharge valve 10 is formed by thinning some portion of the lid 6 through press-forming. When a pressure increases in the battery container, the gas discharge valve 10 cleaves, and gas is discharged from the inside of the battery container to reduce the pressure in the battery container, such that the safety of the rectangular secondary battery 20 is ensured.

[0027] In addition, a liquid injection port 9 is formed on the lid 6, and after an electrolyte is injected from the liquid injection port 9 into the battery case 1, a liquid injection plug 11 is welded to the liquid injection port 9, so as to hermetically seal the rectangular secondary battery 20. As the electrolyte, for example, a non-aqueous electrolyte in which a lithium salt, such as lithium hexafluorophosphate (LiPF_6), is dissolved in a carbonate ester-based organic solvent, such as ethylene carbonate, may be used.

[0028] The battery case 1 of the rectangular secondary battery 20 houses the electrode group 3 via an insulating case 2. The insulating case 2 is made of an insulating resin, such as polypropylene, for example, and electrically insulates the battery case 1 from the electrode group 3.

[0029] As shown in FIG. 3, the electrode group 3 is a wound electrode group in which a positive electrode body (positive electrode) 301 and a negative electrode body (negative electrode) 302 are wound about a winding axis L in a flat shape with a separator 303 interposed therebetween. The electrode group 3 includes a positive electrode foil-exposed portion 301c and a negative electrode foil-exposed portion 302c serving as metal-foil-exposed portions (described later), which are at least partially formed in a flat plate shape, as shown in FIG. 2. Then, the flat-shaped portions of the positive electrode foil-exposed portion 301c and the negative electrode foil-exposed portion 302c are respectively overlaid with and connected to one end of a positive electrode collector plate 4A and one end of a negative electrode collector plate 4B by welding or the like. It should be noted that FIG. 3 shows a state of the electrode group 3 before the metal-foil-exposed portions (the positive electrode foil-exposed portion 301c, the negative electrode foil-exposed portion 302c) of the positive electrode body 301 and the negative electrode body 302 are formed in a flat plate shape and bonded to the positive electrode collector plate 4A and the negative electrode collector plate 4B. The electrode group 3 is inserted into the battery case 1 such that the winding axis L direction

of the electrode group **3** is along the longitudinal direction of the battery case **1**.

[0030] The other end (upper end) of the positive electrode collector plate **4A** and the other end (upper end) of the negative electrode collector plate **4B** are electrically connected to the positive electrode external terminal **8A** and the negative electrode external terminal **8B**, respectively. It should be noted that gaskets **5** are interposed between the positive electrode external terminal **8A** and negative electrode external terminal **8B** and the lid **6**, and insulating plates (not shown) are interposed between the positive electrode collector plate **4A** and negative electrode collector plate **4B** and the lid **6**. The gaskets **5** and the insulating plates (not shown) electrically insulate the positive electrode external terminal **8A** and negative electrode external terminal **8B** and the positive electrode collector plate **4A** and negative electrode collector plate **4B**, from the lid **6**, respectively.

[0031] The battery case **1**, the lid **6**, the positive electrode collector plate **4A**, and the positive electrode external terminal **8A** are made of aluminum or an aluminum alloy, and the negative electrode collector plate **4B** and the negative electrode external terminal **8B** are made of copper or a copper alloy.

[0032] As shown in FIG. **3**, the electrode group **3** is formed by winding in a flat shape the positive electrode body **301** and the negative electrode body **302** about the winding axis L with the separator **303** interposed therebetween. Here, the positive electrode body **301** includes a positive electrode foil (metal foil) **301a** made of an aluminum foil and positive electrode mixture layers **301b** formed on opposite surfaces of the positive electrode foil **301a**. The end portion of the positive electrode foil **301a** on one side in the width direction (winding axis L direction) includes the positive electrode foil-exposed portion **301c** on which the positive electrode mixture layers **301b** are not provided and the positive electrode foil **301a** is exposed. The negative electrode body **302** includes a negative electrode foil **302a** made of a copper foil and negative electrode mixture layers **302b** formed on opposite surfaces of the negative electrode foil **302a**. The end portion of the negative electrode foil **302a** on the other side in the width direction (winding axis L direction) includes the negative electrode foil-exposed portion **302c** on which the negative electrode mixture layers **302b** are not provided and the negative electrode foil **302a** is exposed. The positive electrode body **301** and the negative electrode body **302** are wound about the winding axis L such that the positive electrode foil-exposed portion **301c** and the negative electrode foil-exposed portion **302c** are disposed on the opposite sides to each other in the winding axis L direction.

[0033] Here, in the present embodiment, each positive electrode mixture layer **301b** includes a first positive electrode mixture layer (first mixture layer) **311** formed on the positive electrode foil **301a** and a second positive electrode mixture layer (second mixture layer) **312** formed on the first positive electrode mixture layer **311** as shown in FIG. **4**. The first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** are configured to have characteristics that are different from each other. Examples of the characteristics that are different from each other may include, but are not particularly limited to, an electrical conductivity, a porosity, and the like. In the thickness direction of the positive electrode mixture layer **301b**, usually the reaction of a positive electrode active material is the largest around the surface of the second positive electrode mixture layer **312** and the reactivity of the positive electrode active material decreases as the distance from the surface of the positive electrode mixture layer **301b** increases (toward the inner part of the positive electrode mixture layer **301b**). Here, if the electrical conductivity of the first positive electrode mixture layer **311** is higher than the electrical conductivity of the second positive electrode mixture layer **312**, for example, the reactivity of the positive electrode active material increases also in the inner part of the positive electrode mixture layer **301b**, thus allowing the positive electrode active material to react more uniformly in the thickness direction of the positive electrode mixture layer **301b**. As a result, it is possible to suppress deterioration of the positive electrode active material of the second positive electrode mixture layer **312** and extend the life of the rectangular secondary battery **20**. In addition, if the porosity of the second positive electrode mixture layer **312** is high, for example, the liquid retention capability for an electrolyte can be

improved.

[0034] In the present embodiment, the first positive electrode mixture layer **311** has a higher electrical conductivity as compared to the second positive electrode mixture layer **312**. In addition, the difference between the L^* value of the first positive electrode mixture layer **311** and the L^* value of the second positive electrode mixture layer **312** is 4 or larger in the $L^*a^*b^*$ color system defined by the International Commission on Illumination (CIE). Further, the first positive electrode mixture layer **311** has a smaller L^* value in the $L^*a^*b^*$ color system as compared to the second positive electrode mixture layer **312**. That is, the first positive electrode mixture layer **311** has a higher black color level as compared to the second positive electrode mixture layer **312**. The reason why the first positive electrode mixture layer **311** has a higher electrical conductivity and a higher black color level as compared to the second positive electrode mixture layer **312** will be described later.

[0035] In addition, in the present embodiment, the width of the first positive electrode mixture layer **311** is greater than that of the second positive electrode mixture layer **312** as shown in FIG. 4 and FIG. 5. Specifically, the end portion of the first positive electrode mixture layer **311** on the one side in the width direction is formed so as to project beyond the second positive electrode mixture layer **312** toward the one side in the width direction. On the other hand, the end portion of the first positive electrode mixture layer **311** on the other side in the width direction is formed so as to be flush with the end portions of the second positive electrode mixture layer **312** and the positive electrode foil **301a** on the other side in the width direction. This means that the width of the first positive electrode mixture layer **311** is equal to the distance from the end portion of the first positive electrode mixture layer **311** on the one side to the end portion of the second positive electrode mixture layer **312** on the other side. Thus, the width of the first positive electrode mixture layer **311** can be measured from above the second positive electrode mixture layer **312** (in the thickness direction of the positive electrode body **301**).

[0036] In addition, in the present embodiment, since the surface of the end portion of the first positive electrode mixture layer **311** on the one side is not covered by the second positive electrode mixture layer **312**, the first positive electrode mixture layer **311** contacts an electrolyte in the end portion of the first positive electrode mixture layer **311** on the one side. At this time, in the end portion of the first positive electrode mixture layer **311** on the one side, the reaction of the positive electrode active material is the largest around the surface of the first positive electrode mixture layer **311** in the thickness direction of the first positive electrode mixture layer **311** and the reactivity of the positive electrode active material decreases as the distance from the surface of the first positive electrode mixture layer **311** increases (toward the inner part of the first positive electrode mixture layer **311**). Here, since the first positive electrode mixture layer **311** has a relatively high content of carbon particles and high electrical conductivity (for example, as compared to the second positive electrode mixture layer **312**), which will be described later, the reactivity of the positive electrode active material is relatively high also in the inner part of the first positive electrode mixture layer **311**. This allows the positive electrode active material to react more uniformly in the thickness direction of the first positive electrode mixture layer **311**, and thus can suppress deterioration of the positive electrode active material of the first positive electrode mixture layer **311**. In this way, it is possible to suppress deterioration of the positive electrode active material in the exposed area (area not covered by the second positive electrode mixture layer **312**) of the first positive electrode mixture layer **311**.

[0037] In addition, in the present embodiment, the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** contain carbon particles as a conductive material, and the average particle diameter of the carbon particles of the first positive electrode mixture layer **311** may be smaller than the average particle diameter of the carbon particles of the second positive electrode mixture layer **312**, as will be described later. Such a configuration increases the specific surface area of the carbon particles of the first positive electrode mixture layer **311**, and thus

increases the contact area between the carbon particles and the positive electrode active material and allows efficient electron transfer between the carbon particles and the positive electrode active material. In this way, while suppressing deterioration of the positive electrode active material in the exposed area of the first positive electrode mixture layer **311**, it is possible to allow efficient electron transfer between the carbon particles and the positive electrode active material.

[0038] Next, a method for manufacturing the rectangular secondary battery **20** of the present embodiment will be described.

[0039] The method for manufacturing the rectangular secondary battery **20** of the present embodiment includes the step of preparing the positive electrode body **301** and the step of preparing the negative electrode body **302**. For the other steps of the method for manufacturing the rectangular secondary battery **20** of the present embodiment, a known manufacturing method can be used. Thus, the description thereof will be omitted herein.

[0040] As shown in FIG. **6**, the step of preparing the positive electrode body **301** includes a first slurry preparing step **S1**, a second slurry preparing step **S2**, a coating step **S3**, a drying step **S4**, an inspecting step **S5**, a pressing step **S6**, and a cutting step **S7**.

[0041] In the first slurry preparing step **S1**, a first positive electrode mixture slurry (first mixture slurry) **311a**, which will form the first positive electrode mixture layer **311**, is prepared by a positive electrode active material, carbon particles as a conductive material, a binder, and a dispersing solvent for dispersing the positive electrode active material and the carbon particles. Although the materials of the positive electrode active material, the carbon particles, the binder, and the dispersing solvent are not particularly limited, for example, 1 to 10 parts by weight of scaly graphite (carbon particles) as a first conductive material, 1 to 10 parts by weight of powdered carbon particles as a second conductive material, and 1 to 10 parts by weight of polyvinylidene fluoride (hereinafter referred to as PVDF) as a binder are added to 100 parts by weight of lithium nickel manganese oxide as a positive electrode active material, and then N-methylpyrrolidone (hereinafter referred to as NMP) as a dispersing solvent is further added thereto and kneaded to prepare the first positive electrode mixture slurry **311a**.

[0042] It should be noted that examples of the powdered carbon particles may include a carbon black, such as acetylene black, ketjen black, or the like. The average particle diameter of graphite used as the first conductive material is 1 μm to 50 μm , and the average particle diameter of carbon black used as the second conductive material is 1 nm to 500 nm. Further, for the positive electrode active material, other than the lithium nickel manganese oxide, a different lithium manganese oxide having a spinel crystal structure, a lithium manganese composite oxide in which the lithium manganese oxide is partially substituted by or doped with a metal element, a lithium cobaltite or a lithium titanate having a layered crystal structure, or a lithium-metal composite oxide in which the lithium cobaltite or lithium titanate is partially substituted by or doped with a metal element, or the like may be used.

[0043] The second slurry preparing step **S2** may be performed in parallel with the first slurry preparing step **S1**, for example. In the second slurry preparing step **S2**, a second positive electrode mixture slurry (second mixture slurry) **312a**, which will form the second positive electrode mixture layer **312**, is prepared by a positive electrode active material, carbon particles as a conductive material, a binder, and a dispersing solvent for dispersing the positive electrode active material and the carbon particles. Examples of the materials of the positive electrode active material, the carbon particles, the binder, and the dispersing solvent may include, but are not particularly limited to, the same materials as those of the first positive electrode mixture slurry **311a**.

[0044] Here, in the present embodiment, the first positive electrode mixture slurry **311a** has a higher content of carbon particles as the second conductive material as compared to the second positive electrode mixture slurry **312a**. Since the carbon particles as the second conductive material are black, the first positive electrode mixture slurry **311a** containing a large amount of carbon particles has a smaller L^* value in the $L^*a^*b^*$ color system as compared to the second positive

electrode mixture slurry **312a**. That is, the first positive electrode mixture slurry **311a** appears dark (black) as compared to the second positive electrode mixture slurry **312a**. As described above, the content of carbon particles of the first positive electrode mixture layer **311** may be, for example, at least 4% by weight higher than the content of carbon particles of the second positive electrode mixture layer **312** so that the L^* value of the first positive electrode mixture layer **311** is smaller than the L^* value of the second positive electrode mixture layer **312** by 4 or larger. In addition, since the first positive electrode mixture layer **311** has a higher content of carbon particles as compared to the second positive electrode mixture layer **312**, the first positive electrode mixture layer **311** has a higher electrical conductivity as compared to the second positive electrode mixture layer **312**. As described above, this allows the positive electrode active material to react more uniformly in the thickness direction of the positive electrode mixture layer **301b**, and thus can suppress deterioration of the positive electrode active material of the positive electrode mixture layer **301b**.

[0045] Furthermore, when the first positive electrode mixture slurry **311a** is configured to appear darker (black) as compared to the second positive electrode mixture slurry **312a**, the average particle diameter of carbon particles of the first positive electrode mixture slurry **311a** may be smaller than the average particle diameter of carbon particles of the second positive electrode mixture slurry **312a**. As the particle diameter of carbon particles decreases, the specific surface area increases and the amount of light absorption increases. Consequently, the mixture layer appears darker (black).

[0046] The coating step **S3**, the drying step **S4**, and the inspecting step **S5** are performed by using a coating/drying apparatus **200** shown in FIG. 7. The coating/drying apparatus **200** includes a feed roller **201**, a plurality of conveying rollers **202**, a die head **210**, a drying furnace **203**, an inspecting device **250**, and a winding roller **204**.

[0047] The feed roller **201** is configured to support and rotate the roll of the positive electrode foil **301a** so as to unwind the roll and feed the strip-like positive electrode foil **301a** from the roll. The plurality of conveying rollers **202** is disposed along a conveying path of the positive electrode foil **301a** to convey the positive electrode foil **301a** fed from the feed roller **201** to the winding roller **204**. The die head **210** is disposed to face a back roller **202a** among the conveying rollers **202** and is disposed with a predetermined distance from the positive electrode foil **301a**. The die head **210** is used to coat the positive electrode foil **301a** with a slurry. The drying furnace **203** is disposed downstream of the die head **210** in the conveying direction (arrow D direction). The inside of the drying furnace **203** is set to a predetermined temperature to heat and dry the slurry on the positive electrode foil **301a**. The inspecting device **250** is disposed downstream of the drying furnace **203** in the conveying direction. The inspecting device **250** is configured to measure the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** on the positive electrode foil **301a**.

[0048] In the coating step **S3**, the first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a** in a stacked state are applied over the positive electrode foil **301a** by using the die head **210**. In the present embodiment, as shown in FIG. 8 and FIG. 9, the die head **210** is configured such that the first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a** can be applied simultaneously. Specifically, the die head **210** includes a first block **211**, a second block **212**, and a shim **213** sandwiched between the first block **211** and the second block **212**.

[0049] A first supply portion **215** to which the first positive electrode mixture slurry **311a** is supplied from a supply device (not shown) and a first ejection port **216** communicating with the first supply portion **215** to eject the first positive electrode mixture slurry **311a** are provided between the first block **211** and the shim **213**. A second supply portion **217** to which the second positive electrode mixture slurry **312a** is supplied from a supply device (not shown) and a second ejection port **218** communicating with the second supply portion **217** to eject the second positive

electrode mixture slurry **312a** are provided between the second block **212** and the shim **213**. The first ejection port **216** and the second ejection port **218** extend in the width direction of the positive electrode foil **301a**.

[0050] The first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a** are ejected from the first ejection port **216** and the second ejection port **218**, respectively, and applied over the positive electrode foil **301a** in a state where the first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a** are overlaid with each other. At this time, a first positive electrode mixture slurry layer (first mixture slurry layer) **311b** is formed on the positive electrode foil **301a** and a second positive electrode mixture slurry layer (second mixture slurry layer) **312b** is formed on the first positive electrode mixture slurry layer **311b**. The first positive electrode mixture slurry layer **311b** is formed to have a thickness of 25 μm to 50 μm , for example, and the second positive electrode mixture slurry layer **312b** is formed to have a thickness of 25 μm to 50 μm , for example.

[0051] Here, since the width of the second ejection port **218** is less than the width of the first ejection port **216**, the second positive electrode mixture slurry layer **312b** is formed such that the width of the second positive electrode mixture slurry layer **312b** is less than the width of the first positive electrode mixture slurry layer **311b**. It should be noted that the first ejection port **216** and the second ejection port **218** are disposed such that their center lines in the width direction correspond to each other. Thus, as shown in FIG. 10, the opposite ends of the first positive electrode mixture slurry layer **311b** in the width direction are formed so as to project outward beyond the opposite ends of the second positive electrode mixture slurry layer **312b** in the width direction. For example, the width of the second positive electrode mixture slurry layer **312b** is at least 1 mm less than the width of the first positive electrode mixture slurry layer **311b**. Then, the end portion of the first positive electrode mixture slurry layer **311b** in the width direction is formed so as to project 0.5 mm outward beyond the end portion of the second positive electrode mixture slurry layer **312b** in the width direction. With such a configuration, it is possible to surely detect the end portion of the first positive electrode mixture layer **311** and the end portion of the second positive electrode mixture layer **312** in the inspecting step S5, which will be described later.

[0052] In the drying step S4, the positive electrode foil **301a**, the first positive electrode mixture slurry layer **311b**, and the second positive electrode mixture slurry layer **312b** pass through the inside of the drying furnace **203**, whereby the first positive electrode mixture slurry layer **311b** and the second positive electrode mixture slurry layer **312b** are heated and dried to form the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**. At this time, the content of carbon particles of the first positive electrode mixture layer **311** may be 10 to 15% by mass, and the content of carbon particles of the second positive electrode mixture layer **312** may be 6 to 10% by mass. In this case, the L^* value of the first positive electrode mixture layer **311** can be smaller than the L^* value of the second positive electrode mixture layer **312** by 4 or larger while allowing the positive electrode mixture layer **301b** of the positive electrode body **301** to exhibit its required performance.

[0053] In the inspecting step S5, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** on the positive electrode foil **301a** are measured by the inspecting device **250**. Specifically, as shown in FIG. 11, the inspecting device **250** includes a color-difference meter **251** and a lighting unit **252** that are disposed at a predetermined distance from the positive electrode foil **301a**. The lighting unit **252** includes, for example, a halogen lamp and the like and irradiates the positive electrode foil **301a** with light in a substantially perpendicular direction. The lighting unit **252** may be provided separately from the color-difference meter **251** or may be included in the color-difference meter **251**.

[0054] The color-difference meter **251** detects light that is emitted from the lighting unit **252** and reflected on the positive electrode foil **301a**, the first positive electrode mixture layer **311**, and the second positive electrode mixture layer **312**. In addition, the color-difference meter **251** converts a

detected image into values in the $L^*a^*b^*$ color system for each pixel. It should be noted that although the resolution of the color-difference meter **251** is not particularly limited, the color-difference meter **251** detects, as a unit pixel, an area of 1 to 2 mm×1 to 2 mm on the surface of the positive electrode foil **301a**, the first positive electrode mixture layer **311**, and the second positive electrode mixture layer **312**.

[0055] In the present embodiment, since the opposite ends of the first positive electrode mixture layer **311** in the width direction project outward beyond the opposite ends of the second positive electrode mixture layer **312** in the width direction, the light reflected on the opposite end portions of the first positive electrode mixture layer **311** in the width direction is received by the color-difference meter **251**. Further, the first positive electrode mixture layer **311** has a smaller L^* value in the $L^*a^*b^*$ color system as compared to the second positive electrode mixture layer **312**. Since there is such a difference in the L^* value (black color level) in the boundary portion between the positive electrode foil **301a** and the first positive electrode mixture layer **311** and in the boundary portion between the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**, the inspecting device **250** can detect the opposite end portions of the first positive electrode mixture layer **311** and the opposite end portions of the second positive electrode mixture layer **312**. Thus, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** can be measured simultaneously.

[0056] In addition, in the present embodiment, by forming the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** such that the difference between the L^* value of the first positive electrode mixture layer **311** and the L^* value of the second positive electrode mixture layer **312** is 4 or larger, the boundary portion between the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** has an efficiently large difference in the L^* value. Thus, the end portions of the second positive electrode mixture layer **312** can be precisely detected. It should be noted that in an experiment conducted by the inventor of the present application, in the first positive electrode mixture layer **311**, which was formed by adding 8 parts by weight of scaly graphite and 5 parts by weight of powdered carbon particles to 100 parts by weight of lithium nickel manganese oxide, the L^* value had a dispersion 30 (i.e., three times of the standard deviation) of less than 2, and the L^* value was 14 to 15. Meanwhile, in the second positive electrode mixture layer **312**, which was formed by adding 7 parts by weight of scaly graphite and 3 parts by weight of powdered carbon particles to 100 parts by weight of lithium nickel manganese oxide, the L^* value had a dispersion 30 of less than 2, and the L^* value was 24 to 26. Since the L^* value of the first positive electrode mixture layer **311** and the L^* value of the second positive electrode mixture layer **312** both have a dispersion 30 of less than 2, the boundary between the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** can be detected as long as the difference between the L^* value of the first positive electrode mixture layer **311** and the L^* value of the second positive electrode mixture layer **312** is 4 or larger.

[0057] In addition, in the present embodiment, since the positive electrode foil **301a** is made of an aluminum foil, the positive electrode foil **301a** has a larger L^* value as compared to the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**. Further, since the first positive electrode mixture layer **311** is formed to have a smaller L^* value as compared to the second positive electrode mixture layer **312**, there is a large difference between the L^* value of the positive electrode foil **301a** and the L^* value of the first positive electrode mixture layer **311**. With such a configuration, it is possible to easily and precisely detect the boundary between the positive electrode foil **301a** and the first positive electrode mixture layer **311**, and thus it is possible to easily and precisely detect the width of the first positive electrode mixture layer **311**.

[0058] After passing through the inspecting device **250**, the positive electrode foil **301a** is conveyed to the winding roller **204** and wound by the winding roller **204**. The roll wound by the

winding roller **204** is placed on the feed roller **201** in a state where the coated surface is reversed, and then the above-described coating step **S3**, drying step **S4**, and inspecting step **S5** are performed again. In this way, the first positive electrode mixture layers **311** and the second positive electrode mixture layers **312** are formed on the opposite surfaces of the positive electrode foil **301a**.

[0059] Next, in the pressing step **S6**, the positive electrode foil **301a** having the first positive electrode mixture layers **311** and the second positive electrode mixture layers **312** formed on the opposite surfaces thereof is passed between a pair of rollers of a roll press (not shown). The rollers are heated to a temperature of 60° C. to 120° C., and the first positive electrode mixture layers **311** and the second positive electrode mixture layers **312** formed on the opposite surfaces of the positive electrode foil **301a** are heated and compressed. Then, a laminate of the second positive electrode mixture layer **312**, the first positive electrode mixture layer **311**, the positive electrode foil **301a**, the first positive electrode mixture layer **311**, and the second positive electrode mixture layer **312** is formed to have a thickness of 50 μm to 100 μm , for example.

[0060] In the cutting step **S7**, the positive electrode foil **301a** having the first positive electrode mixture layers **311** and the second positive electrode mixture layers **312** formed on the opposite surfaces thereof is cut at its center in the width direction, whereby the positive electrode body **301** shown in FIG. 5 can be obtained.

[0061] It should be noted that although an example of measuring the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** before the cutting step **S7** as shown in FIG. 10 has been described, it is also possible to measure the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** after the cutting step **S7** as shown in FIG. 5. In the cutting step **S7**, the end portion of the first positive electrode mixture layer **311** on the other side in the width direction is formed so as to be flush with the second positive electrode mixture layer **312** and the positive electrode foil **301a**. Meanwhile, the end portion of the first positive electrode mixture layer **311** on the one side in the width direction is formed so as to project beyond the second positive electrode mixture layer **312** toward the one side in the width direction. This means that the width of the first positive electrode mixture layer **311** is equal to the distance from the end portion of the first positive electrode mixture layer **311** on the one side to the end portion of the second positive electrode mixture layer **312** on the other side, as described above. Thus, with the inspecting device **250**, the width of the first positive electrode mixture layer **311** can be measured by detecting the end portion of the first positive electrode mixture layer **311** on the one side and the end portion of the second positive electrode mixture layer **312** on the other side.

[0062] The step of preparing the negative electrode body **302** is a simplified step of preparing the positive electrode body **301**. Specifically, the step of preparing the negative electrode body **302** includes a slurry preparing step, a coating step **S3**, a drying step **S4**, an inspecting step **S5**, a pressing step **S6**, and a cutting step **S7**.

[0063] In the slurry preparing step, a negative electrode mixture slurry, which will form the negative electrode mixture layer **302b**, is prepared by a negative electrode active material, a binder, and a dispersing solvent. Although the materials of the negative electrode active material, the binder, and the dispersing solvent are not particularly limited, for example, 10 parts by weight of PVDF as a binder is added to 100 parts by weight of amorphous carbon powder as a negative electrode active material, and then NMP as a dispersing solvent is further added thereto and kneaded to prepare the negative electrode mixture slurry.

[0064] It should be noted that other than the amorphous carbon, examples of the negative electrode active material may include natural graphite into/from which lithium ions can be insert/removed, or various artificial graphite materials, carbon particles such as coke, or a compound of Si or Sn, and the like (for example, SiO, TiSi.sub.2, and the like), or a composite material of them. Examples of the particle form of the negative electrode active material may include, but are not particularly limited to, a scaly form, a spherical form, a fibrous form, a massive form, or the like.

[0065] In addition, when various types of graphite are used as the negative electrode active material, water can be selected as a dispersing solvent, which can reduce environmental loads. Examples of the water-based negative electrode slurry may include a negative electrode mixture slurry prepared by adding 1 part by weight of styrene butadiene rubber as a binder and 1 part by weight of sodium carboxymethylcellulose as a thickener to 100 parts by weight of natural graphite powder.

[0066] It should be noted that although an example using PVDF as a binder when preparing the negative electrode body **302** and the positive electrode body **301** has been shown, examples of the binder may include a polymer, a mixture, and the like of polytetrafluoroethylene (PTFE), polyethylene, polystyrene, polybutadiene, butyl rubber, nitrile rubber, styrene butadiene rubber, polysulfide rubber, nitrocellulose, cyanoethyl cellulose, various kinds of latex, acrylonitrile, vinyl fluoride, vinylidene fluoride, propylene fluoride, chloroprene fluoride, acrylic-based resin, and the like.

[0067] The coating step **S3**, the drying step **S4**, and the inspecting step **S5** are performed by using the coating/drying apparatus **200** in the same manner as the above-described step of preparing the positive electrode body **301**.

[0068] In this coating step **S3**, however, only one type of negative electrode mixture slurry is applied. Thus, a die head provided with only one ejection port is used. In the coating step **S3**, the negative electrode mixture slurry is applied over the negative electrode foil **302a**.

[0069] In the drying step **S4**, the negative electrode foil **302a** and the negative electrode mixture slurry layer pass through the inside of the drying furnace **203**, whereby the negative electrode mixture slurry layer is heated and dried to form the negative electrode mixture layer **302b**.

[0070] In the inspecting step **S5**, the width of the negative electrode mixture layer **302b** on the negative electrode foil **302a** is measured by the inspecting device **250**.

[0071] After passing through the inspecting device **250**, the negative electrode foil **302a** is conveyed to the winding roller **204** and wound by the winding roller **204**. The roll wound by the winding roller **204** is placed on the feed roller **201** in a state where the coated surface is reversed, and then the above-described coating step **S3**, drying step **S4**, and inspecting step **S5** are performed again. In this way, the negative electrode mixture layers **302b** are formed on the opposite surfaces of the negative electrode foil **302a**.

[0072] Thereafter, in the same manner as the step of preparing the positive electrode body **301**, the pressing step **S6** and the cutting step **S7** are performed, whereby the negative electrode body **302** shown in FIG. **4** can be obtained.

[0073] In the present embodiment, as described above, the width of the first positive electrode mixture layer **311** is greater than the width of the second positive electrode mixture layer **312**. This allows the color-difference meter **251** to detect the light reflected on the opposite end portions of the first positive electrode mixture layer **311** in the width direction and the light reflected on the opposite end portions of the second positive electrode mixture layer **312** in the width direction simultaneously. In addition, the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** have different contents of carbon particles and different L^* values in the $L^*a^*b^*$ color system. Since there is such a difference in the L^* value in the boundary portion between the positive electrode foil **301a** and the first positive electrode mixture layer **311** and in the boundary portion between the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**, the inspecting device **250** including the color-difference meter **251** can detect the end portion of the first positive electrode mixture layer **311** on the one side and the end portion of the second positive electrode mixture layer **312** on the other side, as well as the opposite end portions of the second positive electrode mixture layer **312**. Thus, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** can be measured simultaneously. Consequently, it is possible to reduce the likelihood that the manufacturing process of the positive electrode body **301** becomes complicated.

[0074] In addition, as described above, the positive electrode mixture layer **301b** of the positive electrode body **301** is configured to have a two-layer structure including the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**, and the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** are formed such that their L^* values differ from each other by 4 or larger. In the positive electrode body **301**, even if the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** are configured to have different contents of carbon particles so that their L^* values differ from each other by 4 or larger, it is possible to allow the positive electrode body **301** to easily exhibit its required performance. Therefore, the electrode for a secondary battery of the present invention can easily be applied to the positive electrode body **301**.

[0075] It should be noted that the present invention is not limited to the aforementioned embodiment, and includes a variety of modifications. For example, although the aforementioned embodiment has been described in detail to clearly illustrate the present invention, the present invention need not include all of the structures described in the embodiment. It is possible to replace a part of a structure of an embodiment with a structure of another embodiment. In addition, it is also possible to add, to a structure of an embodiment, a structure of another embodiment. Further, it is also possible to, for a part of a structure of each embodiment, add, remove, or substitute a structure of another embodiment.

[0076] For example, although the aforementioned embodiment shows an example using the same die head **210** to apply the first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a** simultaneously, the present invention is not limited thereto. For example, a separate die head for applying the second positive electrode mixture slurry **312a** may be provided downstream of the die head for applying the first positive electrode mixture slurry **311a** in the conveying direction (arrow D direction), so as to first apply the first positive electrode mixture slurry **311a**, and then apply the second positive electrode mixture slurry **312a**. Then, after the step of drying the first positive electrode mixture slurry **311a** and the second positive electrode mixture slurry **312a**, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** may be measured simultaneously.

[0077] In addition, although the aforementioned embodiment shows an example in which the L^* value of the first positive electrode mixture layer **311** is smaller than the L^* value of the second positive electrode mixture layer **312**, the present invention is not limited thereto. The L^* value of the first positive electrode mixture layer **311** may be larger than the L^* value of the second positive electrode mixture layer **312**. In this case as well, since there is a difference in the L^* value (black color level) in the boundary portion between the positive electrode foil **301a** and the first positive electrode mixture layer **311** and in the boundary portion between the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312**, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** can be measured simultaneously.

[0078] In addition, although the aforementioned embodiment shows an example in which the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** have different contents of carbon particles and different average particle diameters of carbon particles so that their L^* values differ from each other, the present invention is not limited thereto. For example, even when only the average particle diameter of carbon particles of the first positive electrode mixture layer **311** is smaller (or larger) than the average particle diameter of carbon particles of the second positive electrode mixture layer **312**, the L^* value of the first positive electrode mixture layer **311** becomes smaller (or larger) than the L^* value of the of the second positive electrode mixture layer **312**.

[0079] In addition, although the aforementioned embodiment shows an example in which the first positive electrode mixture layer **311** has a different content of carbon particles and the second positive electrode mixture layer **312** has a different average particle diameter of carbon particles,

the present invention is not limited thereto. For example, the content of carbon particles of the first positive electrode mixture layer **311** may be higher (or lower) than the content of carbon particles of the second positive electrode mixture layer **312** and the thickness of the first positive electrode mixture layer **311** may be larger (or smaller) than the thickness of the second positive electrode mixture layer **312**. When the thickness of the first positive electrode mixture layer **311** is larger (or smaller) than the thickness of the second positive electrode mixture layer **312**, the amount of light absorption of the first positive electrode mixture layer **311** becomes larger (or smaller) than the amount of light absorption of the second positive electrode mixture layer **312**. Thus, the L^* value of the first positive electrode mixture layer **311** becomes smaller (or larger) than the L^* value of the second positive electrode mixture layer **312**.

[0080] In addition, the aforementioned embodiment shows an example of irradiating the positive electrode foil **301a** with light from the lighting unit **252** in a substantially perpendicular direction, detecting the light reflected on the positive electrode foil **301a**, the first positive electrode mixture layer **311**, and the second positive electrode mixture layer **312** by the color-difference meter **251**, and measuring the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** by using the L^* value (black color level). In this method, the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** are configured such that their L^* values differ from each other. However, the present invention is not limited thereto. Even when the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** have the same L^* value, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** can be measured simultaneously.

[0081] Specifically, like the inspecting device **250** according to a modification of the present invention shown in FIG. **12**, the lighting units **252** are disposed on the opposite sides in the width direction to irradiate the positive electrode foil **301a** with light in oblique directions. Examples of such a lighting unit **252** may include a ring lighting unit, a bar lighting unit, and the like. Irradiating the positive electrode foil **301a** with light in the oblique directions makes the whole inspection area dark (this may also be referred to as dark field inspection). However, a scattering of the light reflected on the end portion of the first positive electrode mixture layer **311** and the end portion of the second positive electrode mixture layer **312** makes only the end portion of the first positive electrode mixture layer **311** and the end portion of the second positive electrode mixture layer **312** appear bright. By detecting such light with the color-difference meter **251**, the width of the first positive electrode mixture layer **311** and the width of the second positive electrode mixture layer **312** can be measured simultaneously. This method may not use the color-difference meter **251**, and may perform binarization on the detected light so that the end portion of the first positive electrode mixture layer **311** and the end portion of the second positive electrode mixture layer **312** can be detected.

[0082] In addition, although the aforementioned embodiment shows an example using the electrode for a secondary battery of the present invention for the positive electrode body **301**, the present invention is not limited thereto. The electrode for a secondary battery of the present invention may be used for the negative electrode body **302**. However, when the negative electrode mixture layer **302b** is configured to have a two-layer structure including the first negative electrode mixture layer and the second negative electrode mixture layer, the first negative electrode mixture layer and the second negative electrode mixture layer are less likely to have a large difference in the L^* value since the negative electrode mixture layer **302b** of the negative electrode body **302** includes a large amount of amorphous carbon powder, that is, black granular material. For this reason, the measurement may appropriately be performed using the method shown in FIG. **12** when the electrode for a secondary battery of the present invention is used for the negative electrode body **302**.

[0083] In addition, although the aforementioned embodiment shows an example using, as an

electrode for a secondary battery, the one in which the end portion of the first positive electrode mixture layer **311** on the other side is formed so as to be flush with the second positive electrode mixture layer **312** and the positive electrode foil **301a** and the end portion of the first positive electrode mixture layer **311** on the one side is formed so as to project beyond the second positive electrode mixture layer **312** toward the one side in the width direction, the present invention is not limited thereto. The one in which the opposite ends of the first positive electrode mixture slurry layer **311b** in the width direction are formed so as to project outward beyond the opposite ends of the second positive electrode mixture slurry layer **312b** in the width direction may be used as an electrode for a secondary battery.

[0084] In addition, although the aforementioned embodiment shows an example of winding the positive electrode body **301** and the negative electrode body **302** with the separator **303** interposed therebetween, the present invention is not limited thereto. The present invention can also be applied to a secondary battery in which the positive electrode bodies **301** and the negative electrode bodies **302** are alternately stacked one on top of the other with the separators **303** interposed therebetween.

[0085] In addition, although the aforementioned embodiment describes the case where the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** have characteristics that are different from each other, the present invention is not limited thereto. For example, the first positive electrode mixture layer **311** and the second positive electrode mixture layer **312** may be configured to have the same components and composition ratio and may have the same characteristics. In this case, since the method using a difference in the L^* value cannot measure the width of the first negative electrode mixture layer and the width of the second negative electrode mixture layer simultaneously, the measurement may be performed using the method shown in FIG. **12**.

Reference Signs List

[0086] **20** Rectangular secondary battery (secondary battery) [0087] **301** Positive electrode body (electrode for secondary battery) [0088] **301a** Positive electrode foil (metal foil) [0089] **301b** Positive electrode mixture layer (mixture layer) [0090] **311** First positive electrode mixture layer (first mixture layer) [0091] **311a** First positive electrode mixture slurry (first mixture slurry) [0092] **311b** First positive electrode mixture slurry layer (first mixture slurry layer) [0093] **312** Second positive electrode mixture layer (second mixture layer) [0094] **312a** Second positive electrode mixture slurry (second mixture slurry) [0095] **312b** Second positive electrode mixture slurry layer (second mixture slurry layer)

Claims

1. An electrode for a secondary battery comprising: a strip-shaped metal foil; and a mixture layer provided on the strip-shaped metal foil, wherein: the mixture layer includes a first mixture layer provided on the strip-shaped metal foil and a second mixture layer provided on the first mixture layer, and in a width direction of the strip-shaped metal foil, an end portion of the first mixture layer is arranged closer to an end portion of the strip-shaped metal foil than an end portion of the second mixture layer, and is exposed.
2. The electrode for a secondary battery according to claim 1, wherein: the first mixture layer and the second mixture layer contains carbon particles, a content of the carbon particles in the first mixture layer is higher than a content of the carbon particles in the second mixture layer.
3. The electrode for a secondary battery according to claim 2, wherein an average particle diameter of the carbon particles of the first mixture layer is smaller than an average particle diameter of the carbon particles of the second mixture layer.
4. The electrode for a secondary battery according to claim 2, wherein the first mixture layer has a smaller L^* value in a $L^*a^*b^*$ color system by 4 or larger as compared to the second mixture layer.
5. The electrode for a secondary battery according to claim 1, wherein the mixture layer is not

provided at one end portion of the strip-shaped metal foil, and the strip-shaped metal foil is exposed.

6. The electrode for a secondary battery according to claim 1, wherein an electrical conductivity of the first mixture layer is higher than an electrical conductivity of the second mixture layer.

7. The electrode for a secondary battery according to claim 1, wherein a porosity of the second mixture layer is higher than a porosity of the first mixture layer.

8. A secondary battery comprising the electrode for a secondary battery according to claim 1 as a positive electrode.
